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Probability and Statistics (MT1013)

Assignment

PREDICTING INTERNET ADVERTISEMENTS USING STATISTICAL METHODS

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Table 1: Summary of individual contributions and workload.

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1 Overview and Description of the Data

1.1 Content

This assignment analyzes the Internet Advertisements dataset stored in `add.csv`. When the file is imported into R with its header preserved, the raw dataset contains 3279 observations and 1560 columns. The leftmost column is only a numbering or index field; therefore, excluding that field, the dataset contains 1559 attributes.

Each row represents one image instance that must be classified as either an advertisement or a non-advertisement. In the raw attribute layout, the variables can be interpreted as follows:

- Column 0: image height in pixels.
- Column 1: image width in pixels.
- Column 2: aspect ratio, computed as width divided by height.
- Column 3: `Local`, a binary indicator provided by the dataset.
- Columns 4 to 1557: additional binary indicators describing image- or URL-related properties.
- Column 1558: the target label with two possible values, `ad.` and `nonad.`

Although the raw dataset is high-dimensional, this report focuses on a reduced and interpretable subset of variables that is consistent with the group's analysis pipeline. The selected variables are `Height`, `Width`, `Aspect_ratio`, `Local`, and `Is_Ad`. These variables are sufficient for building a statistical classification model while keeping the later analysis transparent and easy to interpret.

1.2 Data Characteristics

The reduced dataset used in this report contains three quantitative variables and two binary variables:

- `Height`, `Width`, and `Aspect_ratio` are quantitative variables that describe the geometry of each image.
- `Local` is a binary indicator.
- `Is_Ad` is the binary response variable, where 1 denotes an advertisement and 0 denotes a non-advertisement.

The target variable is notably imbalanced. In the raw data, 2820 observations belong to the non-advertisement class and 459 observations belong to the advertisement class. Hence, non-advertisements account for approximately 86.0% of the dataset, while advertisements account for only 14.0%. This imbalance is important because it affects how model performance should be evaluated. In particular, accuracy alone is not sufficient, and additional metrics such as recall, specificity, F1-score, and AUC are needed to assess classification quality more reliably.

Another important characteristic is that the dataset is both sparse and heterogeneous. Most of the original attributes are binary indicators, while the three geometric variables vary over wide numeric ranges. This suggests that image instances in the dataset can differ substantially in size and shape, which motivates both descriptive and inferential analyses in the later sections.

1.3 Missing Data

Missing values in the raw CSV file are encoded by the symbol `?` rather than by R's native `NA`. The missingness is concentrated in the geometric variables `Height`, `Width`, and `Aspect_ratio`, while `Local` contains only a small number of missing observations and the response variable `Is_Ad` is complete.

This pattern is important for two reasons. First, the explanatory variables must be standardized and imputed before any model can be fitted. Second, since `Aspect_ratio` is derived from width and height, it is more coherent to recalculate that quantity after imputing the underlying dimensions instead of treating it as an independent raw measurement. The exact missing-value counts and the full cleaning procedure are reported in Chapter 3.

2 Theoretical Basis

2.1 Confusion Matrix

In binary classification, a confusion matrix is a table that compares predicted labels with actual labels. It is a standard tool for evaluating whether a model correctly distinguishes the positive and negative classes. In this report, the positive class corresponds to advertisements, that is, `Is_Ad = 1`.

For a binary classifier, the confusion matrix has the following structure:

Table 2: Structure of a binary confusion matrix

	Predicted Positive	**Predicted Negative**
Actual Positive	True Positive (TP)	False Negative (FN)
Actual Negative	False Positive (FP)	True Negative (TN)

The four entries in the matrix are interpreted as follows:

- True Positive (TP): advertisements that are correctly predicted as advertisements.
- True Negative (TN): non-advertisements that are correctly classified as non-advertisements.
- False Positive (FP): non-advertisements that are incorrectly predicted as advertisements.
- False Negative (FN): advertisements that are incorrectly predicted as non-advertisements.

From these four quantities, several important evaluation metrics can be calculated:

1. **Accuracy**
   Accuracy measures the overall fraction of correct predictions across all instances. While it provides a general sense of model performance, it can be highly misleading when applied to imbalanced datasets.
   $$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$
2. **Precision (Positive Predictive Value)**
   Precision evaluates the reliability of positive predictions. It answers the question: out of all instances predicted as positive, how many were actually positive?
   $$\text{Precision} = \frac{TP}{TP + FP}$$

3. **Negative Predictive Value (NPV)**
   NPV is the negative counterpart to precision, measuring the proportion of correct negative predictions.
$$\text{NPV} = \frac{TN}{TN + FN}$$

4. **Recall (Sensitivity or True Positive Rate)**
   Recall quantifies the model's ability to identify all actual positive cases. There is generally a trade-off between maximizing recall and maximizing precision.
$$\text{Recall} = \frac{TP}{TP + FN}$$

5. **Specificity (True Negative Rate)**
   Specificity assesses the model's effectiveness in correctly rejecting actual negative cases.
$$\text{Specificity} = \frac{TN}{TN + FP}$$

6. **F1-Score**
   The F1-Score calculates the harmonic mean of precision and recall. It is highly valuable for evaluating models on imbalanced datasets, as it ensures both false positives and false negatives are kept in check.
$$\text{F1-Score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

7. **Balanced Accuracy**
   To address the limitations of standard accuracy on skewed data, Balanced Accuracy computes the arithmetic mean of sensitivity and specificity. This prevents the evaluation metric from being artificially inflated by the majority class.
$$\text{Balanced Accuracy} = \frac{\text{Recall} + \text{Specificity}}{2}$$

8. **Receiver Operating Characteristic (ROC) Curve**
   The ROC curve provides a visual representation of a binary classifier's diagnostic ability across varying decision thresholds. It is constructed by plotting the True Positive Rate (TPR) against the False Positive Rate (FPR):

   - **Y-axis (TPR/Recall):** The proportion of actual positives correctly identified.
$$\text{TPR} = \frac{TP}{TP + FN} \tag{1}$$

   - **X-axis (FPR):** The proportion of actual negatives incorrectly classified as positive, equivalent to $1 - \text{Specificity}$.
$$\text{FPR} = \frac{FP}{TN + FP} \tag{2}$$

   An ideal classifier yields an ROC curve that reaches the top-left coordinate $(0, 1)$, indicating perfect sensitivity and zero false positives. A completely random model follows the diagonal reference line from $(0, 0)$ to $(1, 1)$.

9. **Area Under the ROC Curve (AUC)**
   The AUC summarizes the ROC curve into a single scalar value representing the model's discriminative capability. It estimates the probability that the classifier will assign a higher score to a randomly chosen positive instance than to a randomly chosen negative one.

   - **AUC = 0.5:** No discriminative power (equivalent to random guessing).
   - **0.5 < AUC < 1.0:** The model possesses the ability to distinguish between classes.
   - **AUC = 1.0:** A theoretically perfect classification model.

   In practical applications, an AUC between 0.7 and 0.8 is considered acceptable, 0.8 to 0.9 is excellent, and values above 0.9 are deemed outstanding.

2.2 Logistic Regression

2.2.1 Overview

Logistic regression is a statistical method used to model the probability of a binary outcome. In this report, the binary response variable is `Is_Ad`, where:

- `Is_Ad = 1` means the image is an advertisement.
- `Is_Ad = 0` means the image is not an advertisement.

Unlike linear regression, logistic regression is specifically designed for binary responses. The response variable must be binary, while the predictor variables may be quantitative or categorical. This makes logistic regression appropriate for the current assignment, where the goal is to estimate the probability that an image belongs to the advertisement class based on selected image characteristics such as `Height`, `Width`, `Aspect_ratio`, and `Local`.

The logistic regression model outputs a probability between 0 and 1. To convert that probability into a class label, a cutoff threshold is chosen. In the group's implementation, the default threshold is 0.5: observations with predicted probability greater than 0.5 are classified as advertisements, and the remaining observations are classified as non-advertisements.

2.2.2 Mathematical Basis

Let Y_i denote the binary response for observation i, where $Y_i \in \{0, 1\}$. Let p_i be the conditional probability that observation i belongs to the positive class:

$$p_i = P(Y_i = 1 \mid x_{i1}, x_{i2}, \ldots, x_{im})$$

Logistic regression models this probability through the logistic function:

$$p_i = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_m x_{im})}}$$

Figure 1 illustrates the familiar sigmoid shape of the logistic function. The curve stays between 0 and 1, becomes steeper around the middle region, and approaches the two horizontal asymptotes as the linear predictor moves toward large negative or positive values.

Equivalently, the model can be written in logit form:

$$\log\left(\frac{p_i}{1 - p_i}\right) = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_m x_{im}$$

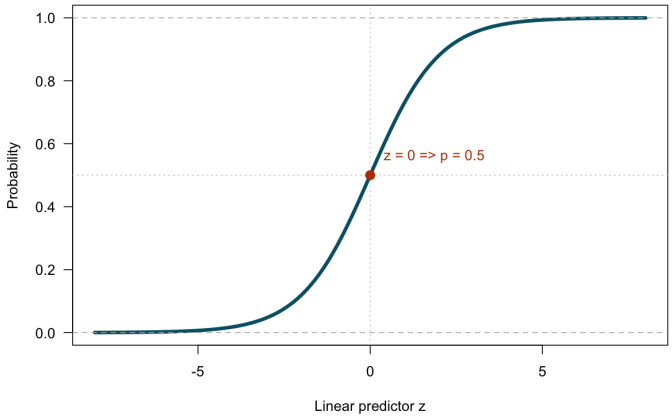


Figure 1: Sigmoid curve used in logistic regression

This transformation is important because it converts a probability in the interval $(0, 1)$ into a linear expression on the real line. In particular, when the linear predictor equals zero, the fitted probability is exactly 0.5, which explains the common default classification threshold used later in this report.

For each predictor x_j, the quantity e^{β_j} is the corresponding odds ratio:

$$e^{\beta_j}$$

If $e^{\beta_j} > 1$, increasing x_j is associated with higher odds that the image is an advertisement. If $e^{\beta_j} < 1$, increasing x_j is associated with lower odds of belonging to the advertisement class.

In the context of this report, logistic regression provides an interpretable framework for assessing how image dimensions and the locality indicator are associated with the probability of an image being an advertisement. This theoretical basis directly supports the model-building and evaluation steps presented in the inferential statistics section.

3 Data Cleaning

3.1 Import, Column Selection, and Basic Cleaning

The data-cleaning stage transforms the raw file `add.csv` into a compact and analysis-ready dataset. The raw CSV contains 3279 observations and 1560 columns, including one numbering column and a large number of sparse binary indicators. Since the later statistical analysis in this project uses only a small subset of interpretable variables, the cleaning script retains the following five fields: `Height`, `Width`, `Aspect_ratio`, `Local`, and `Is_Ad`.

The corresponding R code is shown below.

```
raw_data <- read.csv(
    file = input_file,
    header = TRUE,
    sep = ",",
    check.names = FALSE,
    stringsAsFactors = FALSE,
```

```
    na.strings = "?"
)

selected_column_indices <- c(2L, 3L, 4L, 5L, 1560L)
data <- raw_data[, selected_column_indices]
colnames(data) <- c(
    "Height", "Width", "Aspect_ratio",
    "Local", "Is_Ad"
)

data[] <- lapply(data, standardize_character_column)
```

Listing 1: Importing the raw file and retaining the selected variables

In the revised script, missing markers are standardized in two stages. The import step already recognizes plain `?` tokens, and a subsequent trimming step removes surrounding whitespace so that padded entries such as `" ?"` are also converted to `NA`. The code also uses explicit column indices to make it clear that the report keeps the first four explanatory variables and the final response column from the original file.

The response variable `Is_Ad` is then validated and converted from text labels to a binary representation, where 1 denotes an advertisement and 0 denotes a non-advertisement. The quantitative predictors are coerced to numeric with checks that prevent unexpected strings from being silently converted into missing values.

```
valid_labels <- c("ad.", "nonad.")
data$Is_Ad <- ifelse(data$Is_Ad == "ad.", 1, 0)

for (col in c("Height", "Width", "Aspect_ratio", "Local")) {
    data[[col]] <- coerce_numeric_checked(data[[col]], col)
}
data$Is_Ad <- as.numeric(data$Is_Ad)
```

Listing 2: Validating labels and converting retained variables

Compared with the baseline sample script, the revised version is more reproducible. It resolves the script directory more reliably, even when the project path contains spaces, accepts optional input and output paths, checks that the raw file has the expected number of columns, validates the target labels, and writes the cleaned dataset to a deterministic output location.

3.2 Exploratory Summary and Missing Values

After the basic transformations, a preliminary summary is generated to inspect the scale of each variable and the extent of missingness. Table 3 reports the summary statistics before imputation.

The initial summary already shows several important characteristics of the dataset. First, the variables `Height`, `Width`, and `Aspect_ratio` contain substantial missingness. Second, the means of the geometric variables are noticeably larger than their medians, suggesting right-skewed distributions. Third, the target variable remains strongly imbalanced, with advertisements representing only a small portion of the full sample.

To quantify the missing-data pattern more explicitly, the script counts the number of `NA` values in each column. The result is summarized in Table 4.

Table 3: Summary statistics before imputation

Variable	Min	Q1	Median	Mean	Q3	Max	NAs
Height	1.00	25.00	51.00	64.02	85.25	640.00	903
Width	1.00	80.00	110.00	155.30	184.00	640.00	901
Aspect_ratio	0.0015	1.0357	2.1020	3.9120	5.3333	60.0000	910
Local	0.00	1.00	1.00	0.7672	1.00	1.00	15
Is_Ad	0.00	0.00	0.00	0.1400	0.00	1.00	0

Table 4: Missing values before imputation

Variable	Missing Count	Percentage
Height	903	27.54%
Width	901	27.48%
Aspect_ratio	910	27.75%
Local	15	0.46%
Is_Ad	0	0.00%

These results confirm that missingness is concentrated almost entirely in the explanatory variables, especially the geometric ones. Since the response variable is complete, the dataset remains suitable for supervised classification after preprocessing. However, imputing the missing explanatory values is necessary before any modeling can be performed.

3.3 Imputation, Feature Recalculation, and Export

The imputation strategy is intentionally simple and consistent with the sample workflow. Missing values in `Height` and `Width` are replaced by the median of each column. This choice is robust to the right-skewed distributions observed in Table 3. For `Local`, the missing values are replaced by the rounded mean, which yields the dominant category value of 1.

```
height_median <- median(data$Height, na.rm = TRUE)
width_median <- median(data$Width, na.rm = TRUE)
local_mode_proxy <- round(mean(data$Local, na.rm = TRUE))

data$Height[is.na(data$Height)] <- height_median
data$Width[is.na(data$Width)] <- width_median
data$Local[is.na(data$Local)] <- local_mode_proxy
data$Aspect_ratio <- round(data$Width / data$Height, 4)

write.csv(data, output_file, row.names = FALSE)
```

Listing 3: Imputing missing values and exporting the cleaned dataset

After imputing `Height` and `Width`, the variable `Aspect_ratio` is recalculated rather than imputed directly. This step is important because `Aspect_ratio` is a derived variable, and recalculating it from the cleaned width and height values guarantees internal consistency in the final dataset.

The cleaned output is saved as `preprocessed.csv`. The resulting file contains 3279 observations, 5 numeric variables, and no remaining missing values. In addition, the script prints the

summary statistics before and after imputation together with the replacement values used for each imputed variable. Therefore, the output of the cleaning stage is both analysis-ready and easy to audit before passing the data to the descriptive and inferential analysis stages.

4 Descriptive Statistics

After preprocessing, the final dataset used in this section contains 3279 observations and 5 variables: Height, Width, Aspect_ratio, Local, and Is_Ad. The first three are numerical variables describing image geometry, while Local and Is_Ad are binary variables. Since no missing values remain in the cleaned dataset, this section provides a descriptive summary of the data before moving to inferential analysis. In addition, Aspect_ratio is interpreted here as the ratio recalculated from the cleaned Width and Height variables during preprocessing.

4.1 Distribution of the target variable

The response variable `Is_Ad` indicates whether an image is an advertisement or not. According to the cleaned dataset, 2820 observations belong to the non-advertisement class, whereas 459 observations belong to the advertisement class. Therefore, the target variable is clearly imbalanced, with non-advertisement observations accounting for the majority of the sample.

Table 5 and Figure 2 summarize this class distribution.

Table 5: Distribution of `Is_Ad`

Is_Ad	Frequency	Percentage
0	2820	86.00
1	459	14.00

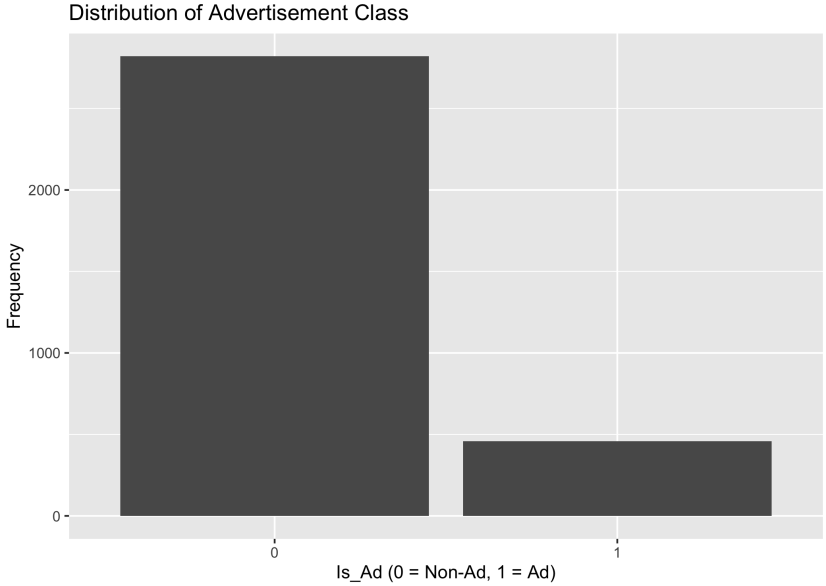


Figure 2: Distribution of Advertisement Class

The bar chart confirms that the response variable is strongly imbalanced. Specifically, non-advertisement images account for 86.00% of the dataset, while advertisement images account for only 14.00%. This imbalance is important because a classifier may achieve high accuracy simply by favoring the majority class. Therefore, later model evaluation should also consider metrics such as recall, specificity, F1-score, and AUC rather than relying on accuracy alone.

4.2 Descriptive statistics of numerical variables

The numerical variables considered in this project are `Height`, `Width`, and `Aspect_ratio`. Their descriptive statistics are reported in Table 6.

In general, the variable `Width` has the largest average value, followed by `Height`, while `Aspect_ratio` measures the proportional relationship between width and height. For all three numerical variables, the maximum values are much larger than the upper quartiles, indicating the presence of extreme observations and outliers. In addition, the means are larger than the medians, which suggests right-skewed distributions.

To complement the numerical summary, the distributions of these variables are visualized using histograms and boxplots in Figures 3 and 4.

Table 6: Descriptive statistics of numerical variables

Variable	Mean	Median	SD	Min	Q1	Q3	Max
Height	60.44	51.00	47.06	1.00	32.50	61.00	640.00
Width	142.89	110.00	112.56	1.00	90.00	144.00	640.00
Aspect_ratio	3.42	2.16	5.20	0.00	1.27	3.90	60.00

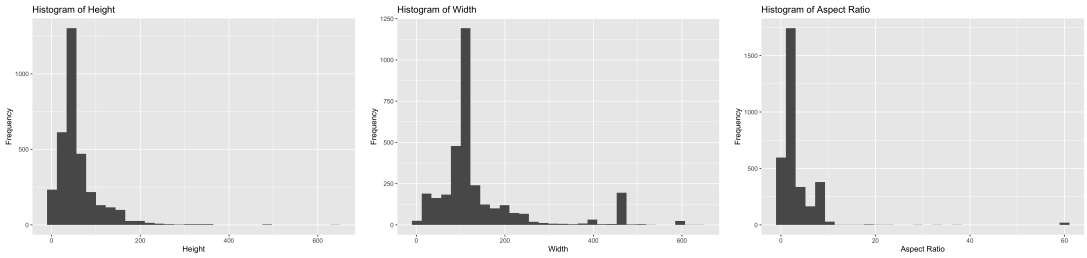


Figure 3: Histograms of Height, Width, and Aspect_ratio

Figure 3 shows that all three numerical variables are positively skewed. Most observations are concentrated in the lower or moderate ranges, while a relatively small number of observations extend far to the right. This pattern is especially clear for `Aspect_ratio`, which has a long right tail.

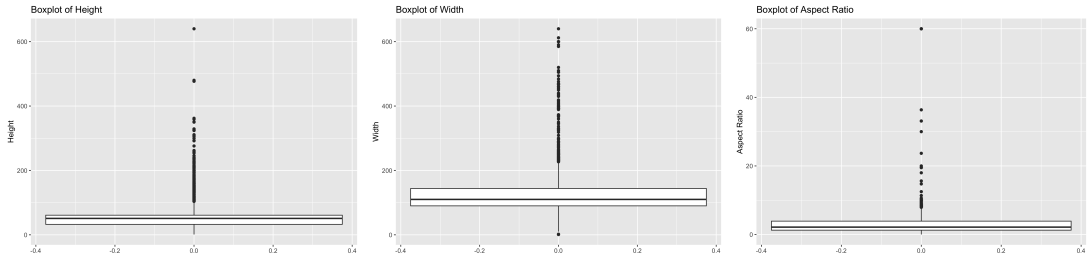


Figure 4: Boxplots of Height, Width, and Aspect_ratio

The boxplots in Figure 4 confirm the presence of many outliers in all three numerical variables. Among them, `Aspect_ratio` exhibits the most pronounced outlier behavior, while `Height` and `Width` also contain several unusually large values.

4.3 Distribution of the Local Variable

The binary variable `Local` is also unevenly distributed. In the cleaned dataset, 760 observations belong to category 0, whereas 2519 observations belong to category 1. Therefore, category 1 accounts for the majority of the dataset.

The distribution is summarized in Table 7 and Figure 5.

Table 7: Distribution of `Local`

Local	Frequency	Percentage
0	760	23.18
1	2519	76.82

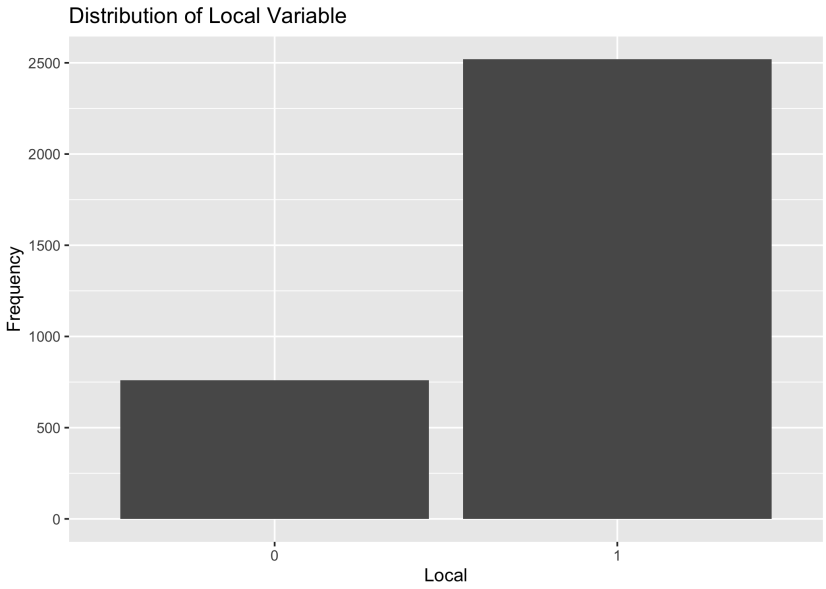


Figure 5: Distribution of Local Variable

The bar chart indicates that `Local = 1` is much more common than `Local = 0`. Hence, the `Local` variable is also imbalanced, although the imbalance is less severe than that of the response variable. This pattern should be kept in mind when interpreting the role of `Local` in the later inferential analysis.

4.4 Comparison by Advertisement Class

To better understand the difference between advertisement and non-advertisement images, the main variables are summarized by `Is_Ad`. Table 8 reports the mean values of the predictors retained in the two classes.

Table 8: Mean values by advertisement class

Is_Ad	Mean Height	Mean Width	Mean Aspect Ratio	Mean Local
0	59.73	118.96	3.16	0.767
1	64.79	289.87	5.07	0.778

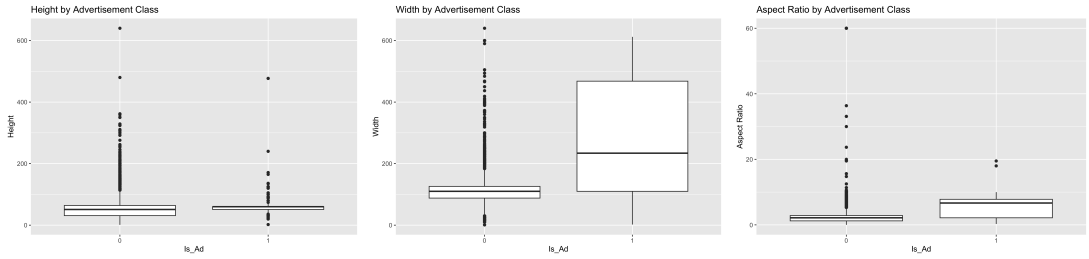


Figure 6: Boxplots of Height, Width, and Aspect_ratio by Advertisement Class

The grouped summaries and boxplots suggest that the clearest differences between the two classes appear in `Width` and `Aspect_ratio`. Advertisement images tend to be much wider than non-advertisement images, with an average width of about 290 compared with about 119 for non-advertisement images. In addition, advertisement images have noticeably larger aspect ratios on average, which is consistent with the idea that many online advertisements appear in wide banner-like formats. The difference in `Height` exists but is less pronounced than the differences observed for `Width` and `Aspect_ratio`.

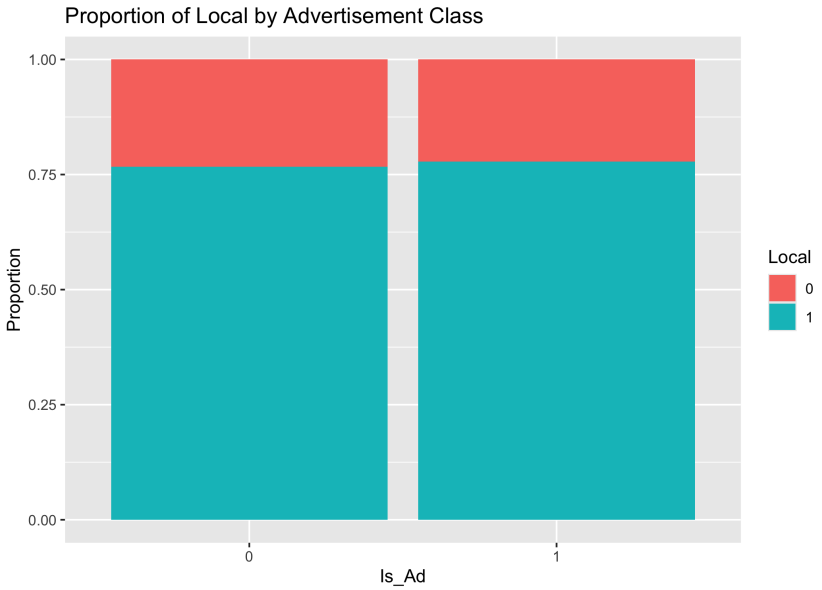


Figure 7: Proportion of Local by Advertisement Class

Figure 7 shows that the proportions of the two `Local` categories are relatively similar across the advertisement and non-advertisement classes. Therefore, compared with `Width` and `Aspect_ratio`, the variable `Local` appears to provide weaker visual separation between the two classes.

4.5 Main findings

Overall, the descriptive analysis highlights three main patterns in the cleaned dataset. First, the response variable is strongly imbalanced, with non-advertisement images accounting for most observations. Second, the numerical variables `Height`, `Width`, and `Aspect_ratio` are all positively skewed and contain outliers, especially `Aspect_ratio`. Third, among the retained predictors, `Width` and `Aspect_ratio` show the clearest differences between advertisement and non-advertisement images, whereas `Local` appears less discriminative.

These descriptive results provide an initial statistical understanding of the cleaned dataset and motivate the inferential analysis in the next section, where the significance of these observed differences will be examined more formally.

5 Inferential Statistics

Following the data preparation phase, we shall apply the Logistic Regression model to the refined dataset in order to evaluate the relationships between variables and formulate comprehensive conclusions based on the empirical evidence.

5.1 Objectives

The purpose of this section is to apply inferential statistical methods to determine which image characteristics are significantly associated with the probability that an internet image is an advertisement. Since the response variable has only two possible outcomes, advertisement

or non-advertisement, logistic regression is an appropriate modeling framework. In addition to predicting the class label, the model allows us to assess the statistical significance, direction, and magnitude of the effect of each predictor. The overall goal is therefore not only to classify observations, but also to draw conclusions about the relationship between the predictors and advertisement status in the broader population.

5.2 Problem statement

Let Y_i denote the response variable for the i-th observation, where:

- $Y_i = 1$ if the image is an advertisement,
- $Y_i = 0$ if the image is a non-advertisement.

The predictors considered in the model are:

- Height
- Width
- Aspect_ratio
- Local

The objective is to model the conditional probability

$$p_i = P(Y_i = 1 \mid x_i)$$

using logistic regression at the significance level $\alpha = 5\%$. The underlying assumptions include independent observations, a sufficiently large sample size, acceptable multicollinearity among predictors, and an approximately linear relationship between the predictors and the log-odds of the response.

5.3 Building the logistic regression model

5.3.1 Training - Testing split

To begin the analysis, the dataset is divided into two subsets: a training set and a test set. The training set is used to fit the logistic regression model, while the test set is used to evaluate the model's performance on unseen data. In this study, the data is split in an 80/20 ratio, where 80% of the observations are assigned to the training set and the remaining 20% to the test set, using the `sample.split` function from the `caTools` library.

```
train_index <- sample.split(ads_data$Is_Ad, SplitRatio = 0.8)

train_set <- subset(ads_data, train_index)
test_set  <- subset(ads_data, !train_index)
```

5.3.2 Fitting logistic models

We then fit a series of models using the training set, where each successive model extends the previous one by including an additional parameter. This stepwise approach allows us to observe how the inclusion of new variables affects the model's performance and helps identify whether the added parameter improves the overall fit. It also provides a clearer basis for comparing models of increasing complexity.

```
mdel1 <- glm(Is_Ad ~ Height + Width,
             data = train_set,
             family = binomial)

mdel2 <- glm(Is_Ad ~ Height + Width + Aspect_ratio,
             data = train_set,
             family = binomial)

mdel3 <- glm(Is_Ad ~ Height + Width + Aspect_ratio + Local,
             data = train_set,
             family = binomial)

summary(mdel1)
summary(mdel2)
summary(mdel3)
```

Model 1: Only Height and Width

Predictor	**Estimate**	**Std. Error**	**z value**	**Pr($>\|z\|$)**
(Intercept)	-3.4624	0.1393	-24.862	$< 2e^{-16}$ ***
Height	-0.0022	0.0014	-1.556	0.12
Width	0.0099	0.0005	20.515	$< 2e^{-16}$ ***
Model Statistics:				
Null deviance: 2123.6		Residual deviance: 1588.4		
AIC: 1594.4		Fisher Scoring iterations: 5		

Table 9: Summary of Model 1 (Height + Width)

- **Key variables:**
  - **Height**: This variable is not statistically significant ($p = 0.12$), suggesting that, by itself, it does not provide strong predictive value.
  - **Width**: This variable is statistically significant ($p < 2 \times 10^{-16}$) and has a positive coefficient, indicating that wider units are more likely to be classified as advertisements.
- **AIC**: 1594.4
- **Residual deviance**: 1588.4

Model 2: Adding `Aspect_ratio`

Predictor	Estimate	Std. Error	z value	Pr($>\|z\|$)
(Intercept)	-2.8945	0.1916	-15.104	$< 2e^{-16}$ ***
Height	-0.0141	0.0025	-5.734	$9.83e^{-09}$ ***
Width	0.0164	0.0011	14.782	$< 2e^{-16}$ ***
Aspect_ratio	-0.2556	0.0515	-4.960	$7.06e^{-07}$ ***
Model Statistics:				
Null deviance: 2123.6		Residual deviance: 1453.4		
AIC: **1461.4**		Fisher Scoring iterations: 6		

Table 10: Summary of Model 2 (Adding Aspect_ratio)

- **Significant variables:**
  - **Height**: After adding `Aspect_ratio`, the effect of `Height` changes noticeably. It becomes highly significant ($p \approx 1 \times 10^{-8}$) with a negative coefficient, indicating that, when width and aspect ratio are held constant, taller units are less likely to be classified as advertisements.
  - **Width**: This variable remains statistically significant.
  - **Aspect_Ratio**: This variable is also significant ($p \approx 7.06 \times 10^{-7}$) and has a negative coefficient.
- **AIC**: 1461.4
- **Residual deviance**: 1453.4

All three predictors are statistically significant in this model. The inclusion of `Aspect_ratio` lowers the AIC by 133.0 points and reduces the residual deviance by 135.0, suggesting a clear improvement in overall model fit.

Model 3: Adding `Local`

Predictor	Estimate	Std. Error	z value	Pr($>\|z\|$)
(Intercept)	-2.9494	0.2398	-12.302	$< 2e^{-16}$ ***
Height	-0.0140	0.0025	-5.657	$1.54e^{-08}$ ***
Width	0.0164	0.0011	14.768	$< 2e^{-16}$ ***
Aspect_ratio	-0.2559	0.0517	-4.954	$7.27e^{-07}$ ***
Local	0.0647	0.1688	0.383	0.702
Model Statistics:				
Null deviance: 2123.6		Residual deviance: 1453.2		
AIC: 1463.2		Fisher Scoring iterations: 6		

Table 11: Summary of Model 3 (Adding Local)

- **Significant variables:** The three core continuous predictors remain statistically significant, whereas `Local` itself is not significant after being added to the model.
- **AIC**: 1463.2

- **Residual deviance**: 1453.2

Including `Local` leads to only a very small decrease in deviance (0.15) while using one additional degree of freedom. Its large p-value (0.70) indicates that this variable does not meaningfully improve the model. Therefore, `Local` provides little additional explanatory power in this specification.

To further evaluate the importance of each predictor, the nested logistic regression models are then compared through an Analysis of Deviance using the Likelihood Ratio Test. This procedure helps determine whether the drop in residual deviance is large enough to justify introducing an extra variable into the model.

```
anova(mdel1, mdel2, test = "Chisq")
anova(mdel2, mdel3, test = "Chisq")
```

Table 12: Analysis of Deviance Table: Model 1 vs Model 2

Model	Resid. Df	Resid. Dev	Df	Deviance	Pr(>Chi)
1: Height + Width	2620	1588.4			
2: ... + Aspect_ratio	2619	1453.4	1	135.04	$< 2.2e^{-16}$ ***

As reported in Table 12, introducing `Aspect_ratio` produces a marked drop in deviance, with $\Delta D = 135.04$ for just one additional degree of freedom. The corresponding p-value is essentially zero ($p < 2.2 \times 10^{-16}$), showing that the extended model fits the data substantially better than Model 1. This provides strong statistical evidence that `Aspect_ratio` contributes meaningful explanatory power. Therefore, it should be kept in the model.

Table 13: Analysis of Deviance Table: Model 2 vs Model 3

Model	Resid. Df	Resid. Dev	Df	Deviance	Pr(>Chi)
1: ... + Aspect_ratio	2619	1453.4			
2: ... + Aspect_ratio + Local	2618	1453.2	1	0.14776	0.7007

In contrast, Table 13 indicates that the addition of `Local` offers almost no improvement to the model. Although the residual deviance decreases slightly, the change is extremely small ($\Delta D = 0.15$) relative to the cost of introducing one more parameter. Moreover, the associated p-value ($p \approx 0.70$) is well above the conventional 5% significance threshold, showing that this reduction is not statistically meaningful. In practical terms, `Local` does not provide sufficient evidence of an additional contribution once the other predictors are already included.

For this reason, keeping `Local` would only make the model more complex without producing a worthwhile gain in fit. It is therefore reasonable to remove this variable and prefer the more parsimonious specification.

Conclusion: Model 2 is selected as the final model for subsequent evaluation.

5.3.3 Model analysis

Before interpreting the final model, we first examine whether multicollinearity exists among the independent variables. This issue arises when two or more predictors are strongly correlated

with each other, which can make the estimated regression coefficients unstable and harder to interpret. In addition, multicollinearity may inflate the standard errors of the estimates, reducing the reliability of statistical inference. To assess whether this problem is present, we use the Variance Inflation Factor (VIF), a commonly applied diagnostic measure available through the `car` package in R.

```
library(car)
vif(best_model)
```

Predictor	VIF
Height	2.136178
Width	3.720394
Aspect_ratio	3.931560

Table 14: Variance Inflation Factors (VIF) for Model Final

The VIF values for all predictors fall within the range of 2 to 5, suggesting a moderate level of correlation among the explanatory variables rather than severe multicollinearity. In particular, the strongest association appears between `Width` and `Aspect_ratio`, which is consistent with the previously observed correlation coefficient of $r = 0.5483$. Although this indicates some overlap in the information captured by these variables, the magnitude is still within an acceptable range for regression analysis. Therefore, there is no strong evidence that multicollinearity is serious enough to distort the model estimates, and all predictors can reasonably be retained in the final model.

To further examine the contribution of each predictor, we can use the `drop1` function, which evaluates the effect of removing each variable from the fitted model one at a time.

We can evaluate the contribution of each predictor by applying the `drop1` function, which assesses how the model fit changes when individual variables are removed.

```
# Check contribution of each variable
drop1(best_model, test = "Chisq")
```

Variable	Df	Deviance	AIC	LRT	Pr(>Chi)
<none>		1453.4	1461.4		
Height	1	1501.2	1507.2	47.84	4.613×10^{-12} ***
Width	1	2086.0	2092.0	632.62	$< 2.2 \times 10^{-16}$ ***
Aspect_ratio	1	1588.4	1594.4	135.04	$< 2.2 \times 10^{-16}$ ***

Table 15: Single Term Deletions (Drop1 Analysis)

The output summarizes a sequence of likelihood-ratio tests in which each predictor is removed from the model one at a time. For each exclusion, the corresponding changes in residual deviance, AIC, the likelihood-ratio test (LRT) statistic, and its p-value are reported.

In all cases, the removal of a predictor results in a notable increase in both residual deviance and AIC, indicating a significant deterioration in model fit. Additionally, all p-values are well

below the 5% significance threshold, providing strong evidence that each predictor contributes meaningfully to the model.

Hence, `Height`, `Width`, and `Aspect_ratio` should all be retained, as excluding any of them would significantly reduce the model's explanatory power.

Furthermore, confidence intervals can be obtained for the intercept and the regression coefficients, and the corresponding estimated odds ratios can be derived for each predictor.

```
# 95\% confidence intervals
confint(best_model)

# Get odd ratios of each predictor
# OR > 1: increases odds of positive outcome
# OR < 1: decreases odds of positive outcome
exp(coef(best_model))
```

Predictor	2.5 %	97.5 %
(Intercept)	-3.2665	-2.5146
Height	-0.0192	-0.0095
Width	0.0143	0.0187
Aspect_ratio	-0.3650	-0.1649

Table 16: Confidence Intervals for Model Coefficients (95%)

The 95% confidence intervals provide plausible ranges for the true population coefficients. The interval for `Width` is entirely positive [0.0143, 0.0187], indicating a positive association with the response variable. In contrast, the intervals for `Height` [-0.0192, -0.0095] and `Aspect_ratio` [-0.3650, -0.1649] lie entirely in the negative domain, reflecting inverse relationships.

Importantly, none of the intervals include zero, implying that all predictors are statistically significant at the 5% level and have non-zero effects on the likelihood that an image is classified as an advertisement.

5.3.4 Evaluating the model

In logistic regression, the estimated coefficients (β) describe how the log-odds of the response variable change with a one-unit increase in each predictor. Because log-odds are not directly intuitive, the coefficients are exponentiated to obtain Odds Ratios (OR), defined as:

$$\text{OR} = e^{\beta}$$

An Odds Ratio greater than 1 indicates a positive association with the likelihood that an image is classified as an advertisement, while an Odds Ratio less than 1 indicates a negative association. The Odds Ratios obtained from the final model are reported in Table 17.

Predictor	Coefficient (β)	Odds Ratio (e^{β})	Impact Direction
(Intercept)	-2.894	0.0553	Baseline
Height	-0.014	0.9860	Negative (Decreases odds)
Width	0.016	1.0165	Positive (Increases odds)
Aspect_ratio	-0.256	0.7744	Negative (Decreases odds)

Table 17: Odds Ratios of Predictors in the Final Model

Based on the results reported in Table 17, the effects of each predictor on the odds of an image being classified as an advertisement (`Is_Ad` = 1) can be interpreted while holding all other variables constant:

- **Width** (OR $\approx$ 1.017): This is the only predictor with an Odds Ratio greater than 1, indicating a positive association with the outcome. Specifically, a one-unit increase in image width leads to an approximate 1.65% increase in the odds of being an advertisement ($(1.0165-1)\times 100\%$). This result suggests that wider images, such as banner-style formats, are more likely to be advertisements.
- **Height** (OR $\approx$ 0.986): The Odds Ratio is slightly below 1, indicating a negative relationship. A one-unit increase in image height is associated with an approximate 1.40% decrease in the odds of being an advertisement ($(1-0.9860)\times 100\%$). This implies that taller images are less likely to be advertisements.
- **Aspect_ratio** (OR $\approx$ 0.774): This variable exhibits a stronger negative effect compared to the other predictors. A one-unit increase in the aspect ratio reduces the odds of being an advertisement by approximately 22.56% ($(1 - 0.7744) \times 100\%$). This indicates that although width alone increases the likelihood of an advertisement, excessively large aspect ratios (relative to height) tend to decrease this likelihood.

The intercept (OR $\approx$ 0.055) corresponds to the baseline odds of the outcome when all predictors are equal to zero. Although this situation is not practically meaningful in this context, the intercept remains an integral component of the model, as it establishes the starting point from which the effects of the predictors are measured.

To assess the model's ability to generalize to unseen data, predictions were generated on a held-out test set ($n = 656$). The predicted probabilities from the logistic regression model were then converted into binary class labels using a threshold of 0.5. Specifically, observations with $P(Y = 1 \mid X) > 0.5$ were classified as advertisements, while the remaining observations were classified as non-advertisements.

Performance metrics were computed using the `confusionMatrix` function from the `caret` package. The resulting confusion matrix is reported in Table 18.

		Reference (Actual)	
		Non-Ad (0)	**Ad (1)**
Prediction	Non-Ad (0)	557 (TN)	50 (FN)
	Ad (1)	7 (FP)	42 (TP)

Table 18: Confusion Matrix (Prediction vs. Reference)

Based on the confusion matrix, the model achieves an overall accuracy of 91.31% (95% CI: 0.8889, 0.9335), with a p-value of 1.975×10^{-5} for the test [Accuracy > NIR], which is well below the 0.05 significance level. This indicates that the model performs significantly better than the No Information Rate (85.98%), corresponding to a naive majority-class prediction strategy.

A clear trade-off is observed between specificity and sensitivity. The model exhibits very high specificity (98.76%), correctly identifying 557 out of 564 non-advertisement images and producing only 7 false positives. However, its sensitivity (recall) is relatively low at 45.65%, as it correctly detects only 42 out of 92 actual advertisements while missing 50 cases.

In terms of predictive reliability, the precision (positive predictive value) is high at 85.71%, indicating that predictions labeled as advertisements are generally accurate. Similarly, the negative predictive value (NPV) is strong at 91.76%, suggesting that the model is effective at identifying non-advertisements. However, this high NPV may be partially driven by the model's tendency to predict the non-advertisement class more frequently.

The balanced accuracy is 72.2%, reflecting moderate performance across both classes, while the F1-score is 0.596, indicating a moderate balance between precision and recall. Overall, the model demonstrates a conservative classification behavior: it prioritizes precision and specificity when predicting advertisements, but at the cost of missing a substantial number of true advertisement instances due to low recall.

In conclusion, the model exhibits a conservative classification behavior, prioritizing correctness when identifying advertisements, as reflected by its high precision and specificity. However, this strict decision pattern results in a substantial number of missed advertisements, as indicated by its relatively low recall.

Although the confusion matrix provides a useful summary of classification performance at a fixed cutoff value of 0.5, it does not fully describe how well the model distinguishes between the two classes across all possible thresholds. Because model performance may vary depending on the chosen cutoff, relying on a single threshold can give only a limited view of predictive ability. To obtain a more complete assessment, we further examine the **Receiver Operating Characteristic (ROC)** curve together with the **Area Under the Curve (AUC)**, which are standard tools for evaluating the discriminative power of a binary classification model.

As illustrated in Figure 8, the **ROC** curve displays the relationship between the True Positive Rate (Sensitivity) and the False Positive Rate ($1 -$ Specificity) over a range of classification thresholds. This allows us to observe how the model balances correctly identifying positive cases against incorrectly classifying negative cases as positive. The diagonal gray line represents the benchmark of a random classifier, for which the **AUC** equals 0.5. Therefore, the farther the **ROC** curve lies above this baseline, the stronger the model's ability to separate the two classes.

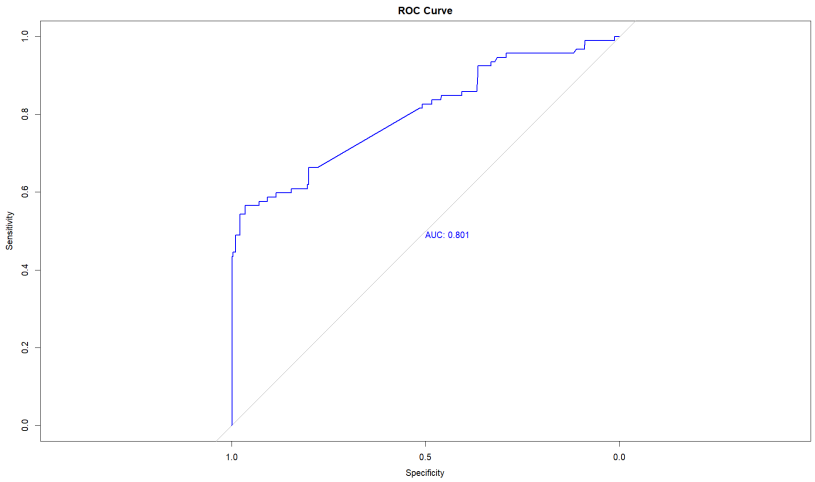


Figure 8: ROC curve for the final model

AUC Interpretation The **Area Under the Curve (AUC)** provides a summary measure of how effectively the model separates the two classes across all possible classification thresholds. A larger AUC indicates stronger discriminative performance, whereas a value close to 0.5 would suggest little better than random guessing.

- **Observed value:** The model achieves an AUC of **0.7783**.
- **Implication:** This value indicates a reasonably strong level of class separation. Because it lies between 0.7 and 0.8, the model can be regarded as having **acceptable to good** discriminatory power. Put differently, when comparing one randomly chosen advertisement with one randomly chosen non-advertisement, the model will assign a higher predicted probability to the advertisement about 77.83% of the time.
- **Shape of the ROC curve:** The plotted curve remains clearly above the diagonal reference line, confirming that the classifier performs substantially better than chance. Its steep rise in the early section suggests that the model can recover a portion of true advertisements while keeping the false positive rate relatively low. However, further gains in sensitivity require increasingly larger sacrifices in specificity, indicating that complete detection of positive cases comes at the cost of more incorrect positive classifications.

6 Discussion and Further Exploration

Based on the inferential statistics and model evaluation performed in the previous sections, we discuss the strengths and limitations of our logistic regression approach and propose directions for future improvements.

6.1 Strengths of the Model

- **Interpretability and Geometric Insight:** The primary strength of this model is its high interpretability. By reducing a massively high-dimensional dataset to just four core predictors (*Height*, *Width*, *Aspect_ratio*, and *Local*), we successfully established a clear profile for advertisements. The analysis revealed that ad status is strongly correlated with

specific dimensions (e.g., wider images being more likely to be ads) and their hosting origin (*Local*).

- **High Specificity (Low False Alarm Rate):** The model achieved an exceptional Specificity of 98.8%. In a practical web browsing context, this is a significant advantage for User Experience (UX). It implies that the model rarely misclassifies legitimate native content as advertisements, meaning users are unlikely to be frustrated by "false alarms" blocking their desired content.
- **Robustness against Multicollinearity:** Despite the inherent mathematical relationship between geometric variables (such as Width, Height, and Aspect Ratio), our VIF analysis confirmed that multicollinearity was kept within acceptable limits ($VIF < 4$), ensuring the stability and reliability of the coefficient estimates.

6.2 Limitations

- **Low Sensitivity (Recall):** The most prominent limitation is the model's low Sensitivity (45.7%). The classifier failed to detect more than half of the actual advertisements in the testing subset (yielding 50 False Negatives against 42 True Positives). This indicates that as a strict "Ad Blocker," the current model would allow many ads to slip through.
- **Class Imbalance Bias:** This low sensitivity is primarily driven by the severe class imbalance inherent in the dataset, where advertisements constitute only roughly 14% of the observations. By attempting to maximize overall accuracy, the model developed a natural bias toward the majority class (non-ads). The default logistic regression decision threshold of 0.5 proved too conservative for capturing the minority class effectively.
- **Linearity Assumption and Imputation Dependency:** Logistic regression assumes a linear relationship between the log-odds and the predictors, which may oversimplify the true behavior of extreme image sizes. Additionally, the necessity of imputing missing values with the median for *Height* and *Width* during the data cleaning phase might have slightly homogenized the data distribution, potentially obscuring boundary cases.

6.3 Recommendations for Further Exploration

To address these limitations and enhance the classification performance, we propose the following extensions for future work:

1. **Threshold Tuning via ROC Analysis:** Instead of strictly adhering to the default probability cutoff of 0.5, we should recalibrate the decision threshold (e.g., lowering it to 0.3 or 0.2). Guided by the ROC curve, lowering this threshold would actively increase Sensitivity (catching more advertisements) at the cost of a marginal decrease in Specificity. This trade-off is highly practical and often necessary for ad detection systems.
2. **Introduction of Interaction Terms:** We recommend adding interaction terms, such as Area = Height × Width. A banner ad (large width, small height) might exhibit a different statistical behavior compared to a pop-up ad that is both wide and tall. Interaction terms would empower the linear model to capture these geometric nuances.
3. **Non-linear Classifiers:** Future iterations should explore non-linear machine learning models, such as Random Forests or Support Vector Machines (SVM). These algorithms are inherently better equipped to handle class imbalances and can delineate the complex

boundaries between advertisements and native content without relying on strict linearity assumptions.

7 Data Source and Code

The complete R source code, processed datasets, and the original raw data are available at the following links:

- **Project Source Code (GitHub):** All R scripts, including data cleaning, model fitting, and visualization code, can be found in our GitHub repository:
  https://github.com/nguyenpham143/BTL-XSTK
  - `add.csv` is the CSV file of the original dataset.
  - `preprocessed.csv` is the dataset after performing data cleaning.
- **Dataset Source (Kaggle):** The original "Internet Advertisements" dataset used for this assignment is sourced from Kaggle:
  https://www.kaggle.com/datasets/uciml/internet-advertisements-data-set/data

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